H01 · S01–S02 (Devore 1.1, 1.3–1.4, 2.1–2.2) — Worked Solutions

Source: Probability and Statistics Homework H01 · study-slides (vetame21.github.io/study-slides)

> [ADDED] The original homework has no solutions. These worked solutions were written for this site and every number was checked in code. Compare them with your lecturer's model answers when those are released.

---

1 Population, sample, and variable (adapted from Devore Ex. 1.3). A quality engineer selects 100 laptops of one brand and model and records, for each, whether it needs service while under warranty.

(a) Identify the population and the sample.

> **Answer:** Population: all laptops of this brand and model. Sample: the 100 selected laptops.

The population is everything the question is about; the sample is the group actually observed.

Why this method: S01: population vs sample.

(b) What is the variable? Categorical or numerical?

> **Answer:** Whether a laptop needs warranty service (yes/no): categorical.

Each laptop gets a label, yes or no. Arithmetic on the labels has no meaning, so it's categorical.

Why this method: Labels → categorical.

(c) If the engineer records the number of service visits instead, is the variable categorical or numerical?

> **Answer:** Numerical (a count: 0, 1, 2, …).

A count is a meaningful amount: 3 visits is more than 1, and differences make sense.

Why this method: Counts → numerical.

(d) Write a question about the population that could be answered from the sample. Descriptive or inferential?

> **Answer:** e.g. "What proportion of ALL laptops of this model need warranty service?": inferential.

It asks about the population using only the sample, so it's inference (which needs a method that accounts for sampling uncertainty). Describing only the 100 sampled laptops would be descriptive.

Why this method: Population claim from a sample → inferential.

2 Sample proportion as a sample mean (adapted from Devore Ex. 1.41). Ten cars crash-tested: S S F S F S S S F S (S = no visible damage).

(a) Find the sample proportion of successes.

> **Answer:** $7/10 = 0.70$

Count the S's: 7 of 10.

Why this method: Proportion = count / total.

(b) Code S = 1, F = 0. Find $\bar{x}$ and compare with (a).

> **Answer:** $\bar{x} = 0.70$, the same as the proportion.

$\bar{x} = \frac{1+1+0+1+0+1+1+1+0+1}{10} = \frac{7}{10} = 0.70$. Adding 0/1 codes just counts the successes, so the mean of 0/1 data IS the proportion of 1s.

Why this method: Mean of an indicator (0/1) variable = proportion.

(c) Ten more cars are tested. How many must be successes so that the proportion in all 20 cars is 0.80?

> **Answer:** 9 successes

Needed total: $0.80 \times 20 = 16$ successes. There are already 7, so $16 - 7 = 9$ of the next 10.

Why this method: Work backwards from the target count.

---

3 Mean, median, and one large value (adapted from Devore Ex. 1.33). Sale prices (thousands of dollars): 590, 820, 570, 610, 350, 1290, 410, 540, 560, 660.

(a) Calculate the sample mean and median.

> **Answer:** $\bar{x} = 640$, $\tilde{x} = 580$ (thousand dollars)

Sum $= 6400$, so $\bar{x} = 6400/10 = 640$. Ordered: 350, 410, 540, 560, **570, 590**, 610, 660, 820, 1290. With $n = 10$ (even), average the two middle values: $\tilde{x} = (570 + 590)/2 = 580$.

Why this method: Mean: add and divide. Median: order first, then take the middle.

(b) If the sixth observation had been 990 instead of 1290, how would the mean and median change?

> **Answer:** Mean drops by 30 to 610; median stays 580.

The sum falls by 300, so the mean falls by $300/10 = 30$. 990 is still the largest value, so the ordered middle pair (570, 590) doesn't change.

Why this method: The mean uses every value; the median uses only position.

(c) Which better describes a typical sale price? Explain.

> **Answer:** The median (580).

One expensive house (1290) pulls the mean up to 640, above 8 of the 10 prices. The median ignores how extreme that value is, so it better represents a typical sale.

Why this method: S01: prefer the median for skewed data or strong extremes.

4 Deviations, variance, standard deviation (adapted from Devore Ex. 1.45, 1.52). Young's modulus (GPa): 118, 114, 113, 116, 114.

(a) Calculate $\bar{x}$ and the five deviations; check they add to 0.

> **Answer:** $\bar{x} = 115$; deviations $3, -1, -2, 1, -1$ (sum 0)

$\bar{x} = 575/5 = 115$. Deviations $x_i - 115$: $3, -1, -2, 1, -1$, and $3 - 1 - 2 + 1 - 1 = 0$ ✓.

Why this method: Deviations about the mean always sum to 0.

(b) Use the deviations to find s^2 and s.

> **Answer:** $s^2 = 4$ GPa², $s = 2$ GPa

$S_{xx} = 9 + 1 + 4 + 1 + 1 = 16$, so $s^2 = \frac{S_{xx}}{n-1} = \frac{16}{4} = 4$ GPa² and $s = \sqrt{4} = 2$ GPa.

Why this method: Square, add, divide by $n - 1$, square root.

(c) Subtract 100 from each observation. Without a full recalculation, state the new variance and explain.

> **Answer:** Still $s^2 = 4$

Shifting every value by the same amount shifts the mean by that amount too, so every deviation is unchanged, and so is the variance.

Why this method: Variance depends only on deviations.

(d) Another sample of $n = 5$ has first four deviations $-1, 2, 0, 3$. What is the fifth?

> **Answer:** -4

All five deviations must sum to 0: $-1 + 2 + 0 + 3 + d_5 = 0 \Rightarrow d_5 = -4$.

Why this method: Use the sum-to-zero property.

5 Outcomes and events (adapted from Devore Ex. 2.4). Each of three mortgages is fixed (F) or variable (V); e.g. FVF.

(a) List the sample space.

> **Answer:** {FFF, FFV, FVF, FVV, VFF, VFV, VVF, VVV}: 8 outcomes

Each outcome is an ordered string of length 3. Hold the first letter at F and list the four endings, then repeat with V.

Why this method: S02: list systematically.

(b) A = exactly two fixed.

> **Answer:** A = {FFV, FVF, VFF}

Exactly one V, in any of the three positions.

Why this method: Select the outcomes that fit.

(c) B = all three the same type.

> **Answer:** B = {FFF, VVV}

Why this method: Select the outcomes that fit.

(d) C = at most one variable.

> **Answer:** C = {FFF, FFV, FVF, VFF}

Zero V's (FFF) or exactly one V.

Why this method: "At most one" = 0 or 1.

(e) Find $B \cup C$ and $B \cap C$.

> **Answer:** B ∪ C = {FFF, FFV, FVF, VFF, VVV}; B ∩ C = {FFF}

Union: in either (FFF listed once). Intersection: in both, and only FFF is.

Why this method: ∪ = or (inclusive), ∩ = both.

(f) Are A and B mutually exclusive?

> **Answer:** Yes: A ∩ B = ∅

No outcome has exactly two F's AND all three the same.

Why this method: Disjoint ⇔ empty intersection.

6 Two traffic signals (adapted from Devore Ex. 2.22). P(stop at 1st) = 0.3, P(stop at 2nd) = 0.6, P(stop at at least one) = 0.7. A = stops at first, B = stops at second.

(a) Write the given probabilities in event notation.

> **Answer:** $P(A) = 0.3,\ P(B) = 0.6,\ P(A \cup B) = 0.7$

"At least one" is the union.

Why this method: Translate words to events.

(b) P(stops at both).

> **Answer:** $P(A \cap B) = 0.2$

General addition rule rearranged: $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.3 + 0.6 - 0.7 = 0.2$.

Why this method: Addition rule, solved for the overlap.

(c) P(first but not second).

> **Answer:** $P(A \cap B') = 0.1$

$P(A) - P(A \cap B) = 0.3 - 0.2 = 0.1$.

Why this method: "A only" region of the Venn diagram.

(d) P(exactly one).

> **Answer:** 0.5

A only (0.1) + B only $(0.6 - 0.2 = 0.4)$ $= 0.5$.

Why this method: Exactly one excludes the overlap.

(e) P(neither).

> **Answer:** 0.3

$1 - P(A \cup B) = 1 - 0.7 = 0.3$.

Why this method: Complement of the union.

7 Accidents by shift (adapted from Devore Ex. 2.20). Day: 10% unsafe, 30% unrelated; Swing: 8%, 22%; Night: 6%, 24%. One report is selected at random.

(a) What are the simple events? Check that their probabilities add to 1.

> **Answer:** 6 simple events (shift × cause); sum = 100%

$0.10 + 0.30 + 0.08 + 0.22 + 0.06 + 0.24 = 1.00$ ✓.

Why this method: Simple events are disjoint and fill the sample space.

(b) P(unsafe conditions).

> **Answer:** 0.24

$0.10 + 0.08 + 0.06 = 0.24$.

Why this method: Add the simple events in the event.

(c) P(not day shift).

> **Answer:** 0.60

$P(\text{day}) = 0.10 + 0.30 = 0.40$, so $1 - 0.40 = 0.60$.

Why this method: Complement rule.

(d) P(night or unsafe). Name the rule.

> **Answer:** 0.48 (general addition rule)

$P(\text{night}) + P(\text{unsafe}) - P(\text{night} \cap \text{unsafe}) = 0.30 + 0.24 - 0.06 = 0.48$.

Why this method: General addition rule (the events overlap).

8 Equally likely setups (adapted from Devore Ex. 2.23). Computers 1–2 are laptops, 3–5 desktops; two are chosen at random, all pairs equally likely.

(a) List all possible pairs. How many outcomes?

> **Answer:** 10 pairs: 12, 13, 14, 15, 23, 24, 25, 34, 35, 45

$\binom{5}{2} = 10$ unordered pairs.

Why this method: List systematically (or count with a combination).

(b) P(both laptops).

> **Answer:** $1/10$

Only {1, 2}.

Why this method: $P = N(A)/N$ *for equally likely outcomes.*

(c) P(both desktops).

> **Answer:** $3/10$

{3,4}, {3,5}, {4,5}.

Why this method: $P = N(A)/N.$

(d) Use the complement rule: P(at least one desktop).

> **Answer:** $9/10$

Opposite of "at least one desktop" is "both laptops": $1 - \frac{1}{10} = \frac{9}{10}$.

Why this method: Complement rule.

(e) P(one of each type).

> **Answer:** $6/10 = 3/5$

$2 \times 3 = 6$ mixed pairs (1 or 2 with 3, 4 or 5).

Why this method: Product rule for the favorable count.

(f) A student says each of the three results (two laptops, two desktops, one of each) has probability 1/3. What's wrong?

> **Answer:** The three results are not equally likely.

They contain 1, 3 and 6 of the 10 equally likely pairs, so their probabilities are 1/10, 3/10 and 6/10. N(A)/N only works when counting equally likely outcomes.

Why this method: S02: equal likelihood is an assumption, not a default.